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REGULAR CONTRIBUTION

A unified drift-adaptive and robust residual autoencoder with attention and few-shot learning for zero-day intrusion detection

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Abstract

Autoencoder-based unsupervised intrusion detection systems (IDS) offer a promising approach for identifying zero-day attacks without requiring labeled malicious data. However, residual autoencoder (RAE) frameworks remain limited by static decision thresholds, susceptibility to concept drift (CD), limited adversarial robustness, and restricted adaptability to emerging attack patterns. To address these limitations, we propose the Attention-Boosted Streaming Residual Autoencoder (ASRAE), a unified framework that integrates attention-enhanced representation learning, online CD detection with adaptive thresholding, benign-only adversarial training, post-identification few-shot meta-learning, and explainability through integrated gradients (IG). Experiments on UNSW-NB15, CIC-IDS2017, and CSE-CIC-IDS2018 show that ASRAE outperforms the evaluated unsupervised baselines, achieving ROC-AUC values of 0.952, 0.940, and 0.931, respectively. Under the controlled simulated concept-drift setting, ASRAE maintains a ROC-AUC above 0.91, while adversarial evaluation using Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD) demonstrates improved robustness to feature-space perturbations. The few-shot adaptation stage further enables adaptation to held-out attack families using only a small number of labeled samples while largely preserving benign reconstruction performance. ASRAE requires an average inference time of 9.2 ms per sample in the evaluated experimental environment. Overall, the results demonstrate the effectiveness of ASRAE across the benchmark, simulated-drift, adversarial, and few-shot adaptation settings.

Keywords Residual autoencoder · Network intrusion detection · Zero-day attack · Concept drift · Adversarial robustness · Few-shot learning · Explainable AI

1 Introduction

The increasing digitalization of modern society has led to the widespread deployment of interconnected systems across critical domains, including healthcare [1], finance [2], transportation [3], industrial control systems [4], and smart cities [5]. While this rapid connectivity has improved efficiency and automation, it has also significantly expanded the attack surface of modern networks, exposing them to increasingly sophisticated cyber threats. Among these, zero day attacks, which exploit previously unseen vulnerabilities, pose a critical challenge to conventional intrusion detection systems (IDS) [6]. Traditional signature based approaches are inherently limited in this context, as they rely on predefined attack patterns and fail to generalize to emerging threats. As a result, anomaly based IDS, particularly those leveraging deep learning, have gained increasing attention due to their ability to model normal network behavior and identify unusual activities without requiring labeled malicious data [7, 8].

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Among anomaly based approaches, deep autoencoder models have become widely used for unsupervised intrusion detection. These models learn compact representations of benign network traffic and detect anomalies by measuring reconstruction errors between the original and reconstructed inputs [6, 9]. Residual autoencoders (RAE) further improve this process by incorporating skip connections and hybrid loss functions, which enhance training stability and reconstruction quality [10]. However, despite these advancements, existing autoencoder based intrusion detection frameworks still face important limitations, particularly in dynamic and adversarial environments where network behavior continuously evolves [11].

First, these frameworks remain inherently static because they are trained only once on historical benign traffic and cannot adapt to concept drift caused by evolving network behavior, newly emerging applications, or temporal traffic variations. Second, anomaly detection thresholds are typically fixed using statistical measures derived from the training set, which may become unreliable over time and lead to increased false positive rates. Third, existing models generally lack robustness against adversarial perturbations, making them vulnerable to evasion attacks due to the absence of dedicated defense mechanisms. In addition, their generalization capability is often limited because evaluation is commonly restricted to a single dataset, leaving their effectiveness across diverse and modern network environments uncertain. Finally, these models operate largely as black boxes and provide limited insight into why specific traffic samples are classified as anomalous, thereby reducing interpretability and complicating forensic analysis.

To address these limitations, we propose the Attention Boosted Streaming Residual Autoencoder (ASRAE), a unified and adaptive framework for robust anomaly detection in dynamic network environments. Unlike existing approaches that treat robustness, adaptability, and interpretability as separate problems, the proposed framework integrates these capabilities within a single architecture. This unified design enables the model to effectively handle evolving data distributions, adversarial perturbations, and emerging attack patterns in a coherent manner.

The contribution of ASRAE lies in the coordinated use of multiple adaptation mechanisms within a common reconstruction-based framework rather than in claiming novelty for each component individually. The attention-enhanced latent representation supports anomaly reconstruction and selective few-shot adaptation, while reconstruction-error statistics are used for both anomaly scoring and online drift monitoring. Benign-only adversarial training further improves robustness without requiring attack labels during initial training. In this way, ASRAE extends the original static residual autoencoder toward a unified framework for robust reconstruction, online adaptation, and post-identification few-shot updating.

The main contributions are summarized as follows:

- *Drift-aware adaptive anomaly detection:* A streaming adaptive threshold based on reconstruction-error statistics is used to respond to changing traffic distributions and support selective online retraining.
- *Attention-enhanced latent representation:* Multi-head self-attention is incorporated into the latent representation to model dependencies among compressed features and improve anomaly discrimination.
- *Benign-only adversarial robustness:* Adversarial perturbations are applied to benign training samples to improve reconstruction stability and reduce sensitivity to evasion-oriented perturbations.
- *Few-shot adaptation for emerging attacks:* MAML-based selective adaptation enables the model to update a restricted set of parameters when only a small number of labeled samples from a newly identified attack family are available.
- *Explainable anomaly detection:* Integrated gradients are used to provide feature-level attribution for detected anomalies and improve interpretation of model decisions.

The framework is evaluated on UNSW-NB15, CIC-IDS2017, and CSE-CIC-IDS2018, achieving ROC-AUC values of 0.952, 0.940, and 0.931, respectively, with a mean ROC-AUC of approximately 0.941. Under the controlled simulated concept-drift setting, ASRAE maintains more stable performance than the original RAE, while the adversarial experiments show improved resistance to feature-space perturbations. The reported inference time remains below 10 ms per sample in the evaluated experimental environment.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the proposed framework. Section 4 describes the experimental setup and results. Section 5 discusses the findings and limitations, and Sect. 6 concludes the paper.

2 Related work

2.1 Unsupervised IDS using autoencoders

Autoencoders have been widely adopted for unsupervised network anomaly detection due to their ability to learn representations of benign traffic without requiring labeled malicious data. Early work by Javaid et al. [12] utilized a sparse autoencoder combined with softmax regression for attack detection, while Farahnakian and Heikkonen [13] proposed a stacked autoencoder to capture hierarchical network

features. Shone et al. [14] further enhanced anomaly discrimination by integrating a non-symmetric deep autoencoder with a random forest classifier.

More recent approaches have focused on improving representation learning through advanced architectures. These include attention-based autoencoders [15], hybrid autoencoder–transformer models [16], and generative adversarial network-based frameworks [17]. Attention-based feature interaction has also been investigated in related representation-learning tasks. C2T-Net employs channel-aware self-attention and cross-fusion to combine complementary feature representations, while CLEAR uses cross-transformer interactions for person attribute recognition and retrieval [18, 19]. Although these methods address computer vision rather than intrusion detection, they provide relevant examples of attention mechanisms for modeling interactions among learned feature representations. The residual autoencoder (RAE) [10] represents a notable advancement by incorporating skip connections and hybrid loss functions to improve training stability and reconstruction fidelity. However, despite these improvements, existing autoencoder-based intrusion detection systems remain largely static and lack adaptability to evolving network conditions, limiting their effectiveness in dynamic environments [20].

2.2 Concept drift and adaptive IDS

Concept drift (CD) significantly affects the performance of intrusion detection systems in dynamic network environments, where data distributions evolve over time. While various drift detection techniques, such as ADaptive WINdowing (ADWIN) and Drift Detection Method (DDM), have been proposed, they are primarily designed for supervised learning settings [21]. Existing unsupervised approaches for drift adaptation remain limited. For example, ExpIDS [22] employs a denoising autoencoder for drift handling, but it does not incorporate residual learning or dynamic thresholding mechanisms. Similarly, [23] propose an adaptive deep autoencoder for security anomaly detection in evolving network environments, but do not address adversarial robustness or few-shot generalization to unseen attacks. Robustness to general distribution shift has also been studied structurally in [24], though not in the context of unsupervised intrusion detection. Deep active learning provides methodologies to select informative and trustworthy unlabelled instances for human labelling [25]. This body of work bears relevance for our online retraining buffer defined in Eq. (10), which currently selects candidate retraining samples purely by reconstruction-error threshold without active sample selection criteria. Separately, emerging dynamic graph continuous state-space models are designed for evolving networked systems [26]. Our ASRAE treats each network flow as an independent tabular vector and does not explicitly model host-level graph interactions across flows; integrating dynamic graph representations constitutes a promising future research direction.

Overall, there is a clear lack of methods that integrate online drift detection, adaptive thresholding, and efficient model updating within a residual autoencoder framework, which remains an open research challenge.

2.3 Adversarial robustness in autoencoders

Adversarial attacks, such as those generated using gradient based perturbation methods, can significantly degrade the performance of autoencoder based intrusion detection systems. Existing defense strategies, including adversarial training, are predominantly designed for supervised classifiers rather than unsupervised models [27]. Some approaches, such as ensemble based defense frameworks [28], incorporate adversarial training in autoencoder based systems, but they rely on both benign and malicious samples during training. Similarly, AIDTF [29] focuses on supervised detection models and does not address unsupervised reconstruction based frameworks.

2.4 Few-shot learning for intrusion detection

Few shot learning has emerged as an effective strategy for addressing data scarcity in intrusion detection systems. However, most existing approaches are limited to supervised or semi supervised settings [30, 31]. For instance, adversarial autoencoder based methods [32] reduce labeling requirements but do not fully operate in an unsupervised manner. Modelagnostic meta learning (MAML) based approaches have also been explored for intrusion detection [33] but they are primarily applied to classification models and not to unsupervised autoencoder based frameworks. Fine-tuning-based adaptation under limited labeled data has similarly been used for defect detection in other domains, such as industrial fabric inspection [34], underscoring that data-efficient adaptation is a cross-domain need rather than one unique to network security. Consequently, enabling rapid adaptation of unsupervised models to new attack types using limited labeled data remains a significant research gap.

2.5 Explainable anomaly detection

Explainable artificial intelligence methods, including SHAP and LIME, have been widely used to improve the transparency of intrusion detection systems [35]. However, their application to autoencoder based models, particularly residual architectures, remains limited. Existing work, such as convGRU based autoencoders, has explored interpretability using layer wise relevance propagation [36], but alternative approaches such as integrated gradients have not been suf-

ficiently investigated in this context. Providing interpretable anomaly detection within residual autoencoder frameworks is therefore an important yet underexplored direction.

Overall, existing studies address adaptability, robustness, generalization, and interpretability in isolation. There remains a lack of a unified framework that simultaneously integrates concept drift adaptation, adversarial robustness, few shot generalization, and explainable anomaly detection within a single unsupervised architecture. This gap motivates the proposed ASRAE framework.

3 Proposed method

3.1 Overall architecture

The proposed ASRAE framework builds upon the core residual autoencoder architecture, which consists of an encoder, decoder, residual skip connection, and a hybrid loss function. To overcome the limitations of existing models, the framework integrates multiple functional components within a unified design to enhance adaptability, robustness, and interpretability. Figure 1 illustrates the overall architecture of ASRAE and its role in the intrusion detection pipeline. The framework follows an end to end processing flow, where each component contributes to effective anomaly detection. The encoder transforms the input network traffic features, represented as a 49 dimensional vector, into a compact latent representation of dimension 32. This transformation reduces computational complexity while preserving essential information.

The latent representation is further processed using a multi head attention mechanism, which captures dependencies among compressed features and highlights subtle patterns that may indicate anomalous behavior. The attention enhanced latent vector is then passed to the decoder, which reconstructs the original input. A residual skip connection is incorporated to preserve important input features and stabilize the training process. The model is optimized using a hybrid loss function that combines Smooth L1 loss and cosine similarity. This formulation ensures accurate reconstruction while maintaining consistency in feature representation, allowing the model to learn meaningful patterns rather than trivial mappings.

In addition to the core architecture, the framework incorporates auxiliary components, including adversarial training, concept drift detection, few shot adaptation, and explainability mechanisms. These components are seamlessly integrated into the overall system to improve performance in dynamic and adversarial environments. As a result, ASRAE retains the efficiency of the residual autoencoder while addressing its key limitations, including limited discriminative capability, vulnerability to adversarial perturbations, lack of adaptability, and absence of interpretability.

3.2 Attention-boosted latent representation

Given a normalized input feature vector $x \in \mathbb{R}^{49}$, the encoder f_ϕ generates a compact latent representation

$$z = f_\phi(x), \qquad z \in \mathbb{R}^{d_{\text{latent}}}, \tag{1}$$

where $d_{\text{latent}} = 32$. To apply multi-head self-attention in a mathematically meaningful manner, the latent representation is organized into $N = 4$ latent feature groups, each containing $d_g = 8$ dimensions, such that

$$Z = \text{reshape}(z, N, d_g) \in \mathbb{R}^{4\times 8}, \qquad N d_g = d_{\text{latent}}. \tag{2}$$

The resulting rows of Z are treated as latent feature tokens rather than temporal or spatial tokens. This grouping does not impose any sequential ordering on the network-flow features, and no positional encoding is used. Instead, it provides a structured representation over which the attention mechanism can learn dependencies among different compressed feature groups produced jointly by the encoder.

For each attention head h, the query, key, and value matrices are computed as

$$Q^{(h)} = Z W_Q^{(h)}, \qquad K^{(h)} = Z W_K^{(h)}, \qquad V^{(h)} = Z W_V^{(h)}, \tag{3}$$

where $W_Q^{(h)}$, $W_K^{(h)}$, and $W_V^{(h)}$ are learnable projection matrices associated with the h-th attention head. The scaled dot-product attention is then defined as

$$\text{Attention}\left(Q^{(h)}, K^{(h)}, V^{(h)}\right) = \text{softmax}\left(\frac{Q^{(h)} K^{(h)T}}{\sqrt{d_k}}\right) V^{(h)}, \tag{4}$$

where d_k denotes the dimensionality of the key representation. Because Z contains multiple latent feature groups, the matrix $Q^{(h)} K^{(h)T}$ represents pairwise relationships among these groups rather than a similarity score for a single latent vector.

The outputs of the four attention heads are concatenated and linearly transformed as

$$Z_{\text{att}} = \text{Concat}\,(\text{head}_1, \text{head}_2, \text{head}_3, \text{head}_4)\, W_O, \tag{5}$$

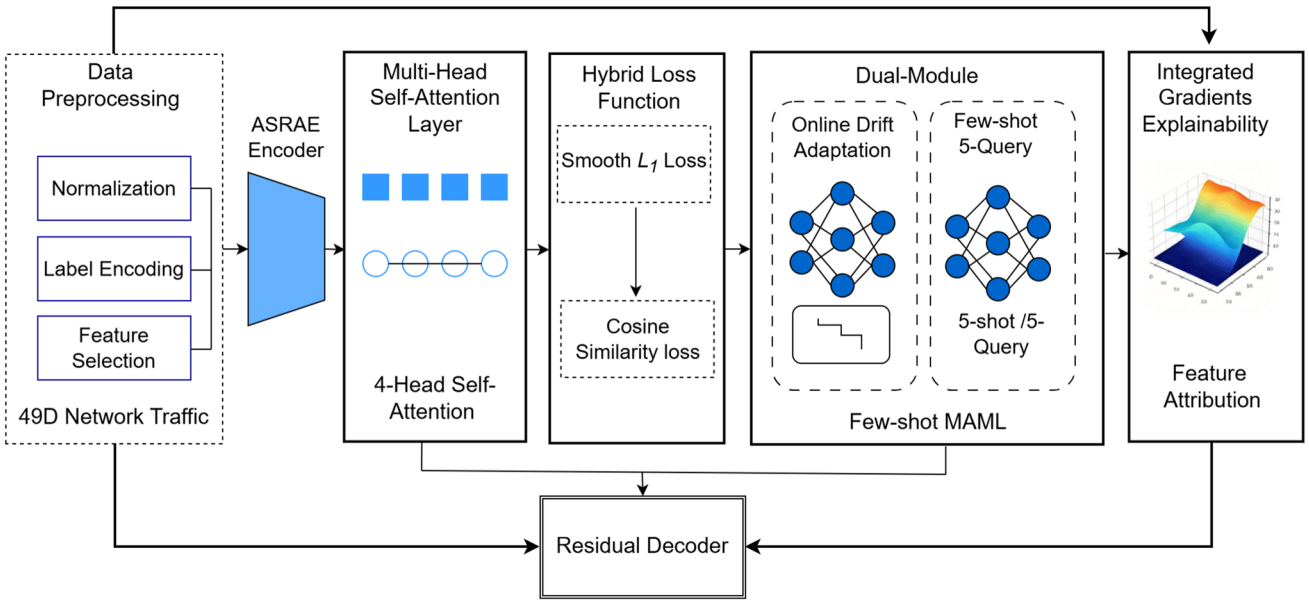


Fig. 1 Architecture of the proposed ASRAE framework. A multi-head attention layer is inserted after the latent representation. The residual connection adds a scaled version of the input to the decoder output. The hybrid loss combines Smooth L1 and cosine similarity

where W_O is the output projection matrix. The attention-enhanced representation is subsequently flattened to recover a d_{latent}-dimensional vector,

$$z_{\text{att}} = \text{vec}\,(Z_{\text{att}}) \in \mathbb{R}^{32}, \tag{6}$$

which is provided to the decoder g_ψ for reconstruction.

The purpose of the attention block is therefore not to model temporal dependencies. Instead, it models relationships among compressed latent feature groups derived from the original network-flow representation. Each attention head can assign different relative importance to interactions among these groups for a given input, allowing the decoder to receive a representation that reflects dependencies that may not be captured as effectively by an attention-free latent mapping. This interpretation is particularly appropriate for tabular network traffic, where dependencies may arise from combinations of multiple flow characteristics even though the input itself has no inherent spatial or sequential structure.

The contribution of this mechanism is also supported by the ablation analysis in Table 9. Removing the attention mechanism from the complete ASRAE framework reduces ROC-AUC from 0.952 to 0.920, while the attention-only configuration achieves a ROC-AUC of 0.924 compared with 0.903 for the original residual autoencoder. These results indicate that the latent attention block contributes additional discriminative capacity beyond the base RAE representation.

3.3 Hybrid loss function

The proposed framework adopts the hybrid loss function from the original RAE to jointly optimize reconstruction accuracy and feature consistency. This loss formulation plays a critical role in guiding the model to learn meaningful representations of network traffic. The loss function is defined as:

$$\mathcal{L} = \alpha \|\hat{x} - x\|_{S1} + (1 - \alpha)(1 - \cos(\hat{x}, x)), \tag{7}$$

where $\|\cdot\|_{S1}$ denotes Smooth L1 loss, $\cos(\hat{x}, x)$ represents the cosine similarity between the input and its reconstruction, and $\alpha = 0.9$ controls the balance between the two terms. The first term focuses on minimizing the numerical reconstruction error, ensuring that the reconstructed output closely matches the original input. The second term enforces consistency in the feature space by preserving the directional similarity between input and reconstruction. This combination encourages the model to capture both fine grained details and global structural patterns, leading to more robust and discriminative representations for anomaly detection.

3.4 Online drift detection and adaptive threshold

During inference, ASRAE maintains a sliding window $\mathcal{W}$ containing the most recent reconstruction errors,

$$e_i = \left\|\hat{x}_i - x_i\right\|_1, \tag{8}$$

with a window size of $W = 1000$. The adaptive anomaly threshold at time step t is defined as

$$T(t) = \mu_t + \beta\sigma_t, \tag{9}$$

where μ_t and σ_t are the mean and standard deviation of the reconstruction errors within $\mathcal{W}$, and $\beta = 3$. A sample is classified as anomalous when $e_i > T(t)$. Updating the threshold from recent reconstruction errors allows the decision boundary to respond to changes in the observed traffic distribution.

A concept-drift event is triggered when five consecutive samples satisfy $e_i > 2T(t)$. The five-sample criterion was selected using the validation data to balance false drift alarms and detection delay. Shorter sequences were more sensitive to transient traffic fluctuations, whereas longer sequences delayed the response to sustained distributional changes.

Following drift detection, partial retraining is performed using the most recent 10% of model-predicted benign samples. In the unsupervised online setting, no ground-truth labels or external oracle are used to verify these samples. Instead, a sample is eligible for the retraining buffer when

$$e_i \leq T(t). \tag{10}$$

Samples with $e_i > T(t)$ are classified as anomalous and excluded from the buffer. However, this procedure cannot guarantee that every retained sample is truly benign. An attack with reconstruction error below the current threshold constitutes a false negative and may therefore enter the retraining buffer. This limitation of unsupervised drift adaptation is further discussed in Sect. 5.

To reduce catastrophic forgetting during partial retraining, elastic weight consolidation (EWC) is applied:

$$\mathcal{L}_{\text{total}} = \mathcal{L} + \frac{\lambda_{\text{EWC}}}{2} \sum_j F_j \left(\theta_j - \theta_j^*\right)^2, \tag{11}$$

where F_j denotes the diagonal Fisher information associated with parameter θ_j, θ_j^* represents the corresponding parameter value before adaptation, and $\lambda_{\text{EWC}} = 1000$. EWC constrains excessive parameter changes during retraining and helps preserve previously learned reconstruction behavior.

3.5 Adversarial training on benign samples

To improve robustness against evasion attacks, the training process incorporates adversarial examples generated from benign samples using the FGSM. The adversarial sample is defined as:

$$x_{\text{adv}} = x + \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}(x, \hat{x})), \tag{12}$$

with perturbation strength $\epsilon = 0.03$. All adversarial perturbations, in both training and evaluation, are constrained to the min–max normalized [0, 1] feature space used throughout preprocessing (Section 4.1); perturbed vectors are clipped to this range after each perturbation step. Perturbations are further restricted to continuous flow-statistic features (e.g., byte counts, duration, packet-length and inter-arrival statistics); categorical and protocol-indicator features are left unperturbed, since arbitrary perturbation of these fields could produce flows that do not correspond to physically realizable network behavior. For datasets requiring dimensionality reduction, adversarial perturbations are generated in the original continuous feature space before applying the fitted PCA transformation. The perturbed samples are then processed using the same transformation applied to clean samples. The model is trained to reconstruct both clean and perturbed benign samples, encouraging the learning of a locally smooth representation in the latent space. This approach enhances robustness without requiring labeled attack data, which is particularly important in unsupervised settings. By focusing on benign samples, the model becomes less sensitive to small input perturbations, thereby reducing vulnerability to adversarial evasion while maintaining stable reconstruction performance.

Equation (12) applies perturbations exclusively to benign training samples and maximises reconstruction loss during model training to enforce latent-space smoothness. This training-time objective differs fundamentally from the behaviour of a real adversarial evasion attacker, whose goal is to reduce the anomaly score of malicious traffic so that attack instances are misclassified as benign. Section 4.8 evaluates practical evasion attacks using this opposite optimisation objective applied to held-out attack test samples.

3.6 Few-shot adaptation for new attack families

ASRAE incorporates model-agnostic meta-learning (MAML) to support rapid adaptation when a small number of labeled samples from a newly identified attack family become available. This stage is separate from the initial unsupervised zero-day detection process, which relies on reconstruction error and the adaptive threshold $T(t)$. Each meta-learning task $\mathcal{T}_i$ contains a support set $\mathcal{S}_i$ and a disjoint query set $\mathcal{Q}_i$. The basic configuration uses $K = 5$ support samples and $Q = 5$ query samples per task. During meta-training, the support set determines the task-specific inner-loop update, whereas the query set is used to evaluate the adapted parameters and optimize the outer meta-objective. To maintain consistency with reconstruction-error-based anomaly detec-

tion, few-shot adaptation uses a label-aware objective. Let $\mathcal{S}_i^b$ and $\mathcal{S}_i^a$ denote the benign and labeled attack samples, respectively, within support set $\mathcal{S}_i$. For a sample x, the reconstruction loss is

$$\mathcal{L}_{\text{rec}}(x) = \alpha \left\| \hat{x} - x \right\|_{\text{S1}} + (1-\alpha)\left[1 - \cos(x, \hat{x})\right], \tag{13}$$

where $\alpha = 0.9$, consistent with the hybrid loss defined in Section 3.3. The reconstruction-error anomaly score is

$$e_\theta(x) = \left\| \hat{x}_\theta - x \right\|_1 . \tag{14}$$

The label-aware support-set objective combines reconstruction preservation for benign traffic with an anomaly-separation term for labeled attacks:

$$\begin{aligned} \mathcal{L}_{\mathcal{S}_i}^{\text{LA}} = & \frac{1}{|\mathcal{S}_i^b|} \sum_{x \in \mathcal{S}_i^b} \mathcal{L}_{\text{rec}}(x) \\ & + \lambda_{\text{sep}} \frac{1}{|\mathcal{S}_i^a|} \sum_{x \in \mathcal{S}_i^a} \max\left(0,\ T(t) - e_\theta(x)\right), \end{aligned} \tag{15}$$

where $\lambda_{\text{sep}} = 1.0$ controls the contribution of the attack-separation term. For a labeled attack whose reconstruction error is already above $T(t)$, the separation term is zero. If its reconstruction error falls below $T(t)$, the term becomes positive and penalizes the model for moving the attack sample into the benign reconstruction region. Consequently, minimizing $\mathcal{L}_{\mathcal{S}_i}^{\text{LA}}$ preserves low reconstruction error for benign samples while maintaining separation between labeled attacks and the current anomaly decision boundary. For task $\mathcal{T}_i$, the task-specific parameters are updated as

$$\theta_i' = \theta - \alpha_{\text{inner}} \nabla_\theta \mathcal{L}_{\mathcal{S}_i}^{\text{LA}}(\theta), \tag{16}$$

where $\alpha_{\text{inner}} = 0.01$. Five inner-loop gradient-update steps are performed for each task. The same label-aware formulation is evaluated on the disjoint query set $\mathcal{Q}_i$ to obtain $\mathcal{L}_{\mathcal{Q}_i}^{\text{LA}}$. The outer meta-objective is

$$\mathcal{L}_{\text{meta}} = \sum_{\mathcal{T}_i} \mathcal{L}_{\mathcal{Q}_i}^{\text{LA}}\left(\theta_i'\right). \tag{17}$$

Thus, the support set determines the task-specific inner-loop adaptation, whereas the disjoint query set determines the outer meta-learning update. The outer-loop learning rate is set to 10^{-3}, with a meta-batch size of 32 and a total of 1200 meta-training episodes. The autoencoder is initially trained exclusively on benign traffic. Worms, Shellcode, and Reconnaissance are withheld from attack-dependent MAML meta-training and are treated as held-out zero-day attack families during meta-testing. Adaptation is performed separately for each held-out family after a small labeled support set becomes available, and performance is evaluated on disjoint samples that are not used during adaptation. Therefore, the reported 5-shot, 10-shot, and 15-shot settings represent post-identification adaptation rather than the initial unsupervised zero-day detection stage. Only the attention projection matrices W_Q, W_K, W_V, and W_O, together with the final decoder layer, are updated during few-shot adaptation. The encoder, the remaining decoder layers, and the residual reconstruction pathway are kept fixed. This restricted update limits the number of adapted parameters and helps preserve the learned benign reconstruction behavior. Consistent with this design, the benign false-positive rate changes from 1.12% before adaptation to no more than 1.15% after 15-shot adaptation, as reported in Table 8.This selective-update principle parallels adapter-based fine-tuning approaches used for few-shot anomaly detection in other domains, where only a lightweight adapter module is updated while a large pre-trained backbone remains frozen [37], reinforcing selective parameter updating as a general-purpose adaptation strategy rather than one specific to network intrusion detection. Few-shot adaptation does not introduce a separate supervised classifier. The final anomaly decision remains reconstruction based:

$$\hat{y}(x) = \begin{cases} 1, & e_{\theta'}(x) > T(t), \\ 0, & e_{\theta'}(x) \le T(t). \end{cases} \tag{18}$$

Therefore, MAML serves as a post-identification adaptation mechanism while preserving the reconstruction-error-based anomaly-detection principle of ASRAE.

We denote detection performance prior to receiving any labelled attack samples as zero-shot performance. Worms, Shellcode, and Reconnaissance undergo strict family-level holdout: these attack families are completely excluded both from autoencoder pre-training and MAML meta-training, instead of relying on simple random sample splitting. To isolate performance gains originating from meta-learning itself, we implement a plain selective fine-tuning baseline which updates exactly the same restricted parameter subset, namely the attention projection matrices W_Q, W_K, W_V, W_O and the final decoder layer. This baseline utilises the identical label-aware loss given in Eq. (15), matching inner-loop learning rate and gradient-step count, but omits the MAML outer-loop meta-training procedure. All reported few-shot results are aggregated across five independent random draws of support-query splits and are presented as mean ± standard deviation in Table 7. Table 8 further compares benign-traffic false-positive rates for both MAML and plain fine-tuning to quantify potential catastrophic-forgetting risk across adaptation strategies.

3.7 Explainability via integrated gradients

To improve interpretability, the proposed framework employs integrated gradients to compute feature attribution scores for each detected anomaly. This approach quantifies the contribution of each input feature to the reconstruction error, enabling transparent analysis of anomalous behavior.

The integrated gradients for the i-th feature are defined as:

$$\mathrm{IG}_i(x) = (x_i - x_i') \times \int_{\alpha=0}^{1} \frac{\partial \mathcal{L}(\hat{x}(\alpha), x(\alpha))}{\partial x_i} d\alpha, \tag{19}$$

where x' denotes a baseline input, set to a zero vector, and the integral is approximated using 50 discrete steps. The loss function L corresponds to the hybrid reconstruction loss used in the model. This mechanism operates in conjunction with the ASRAE architecture by leveraging the reconstruction error to identify the most influential features contributing to each anomaly. The resulting attribution scores can be visualized as heatmaps, highlighting key traffic characteristics associated with anomalous events.

By providing feature level explanations, this module enhances the transparency of the model and addresses the black box limitation commonly associated with deep learning based intrusion detection systems. This capability supports interpretation of individual anomaly decisions and can assist analysts in examining the traffic features associated with detected anomalies.

Figure 2 summarizes the ASRAE workflow from preprocessing and anomaly detection to drift handling and explanation. When concept drift is detected, partial retraining is performed using EWC. Few-shot adaptation is activated separately only when labeled samples from a newly identified attack family become available. Integrated gradients provide feature-level attribution for detected anomalies.

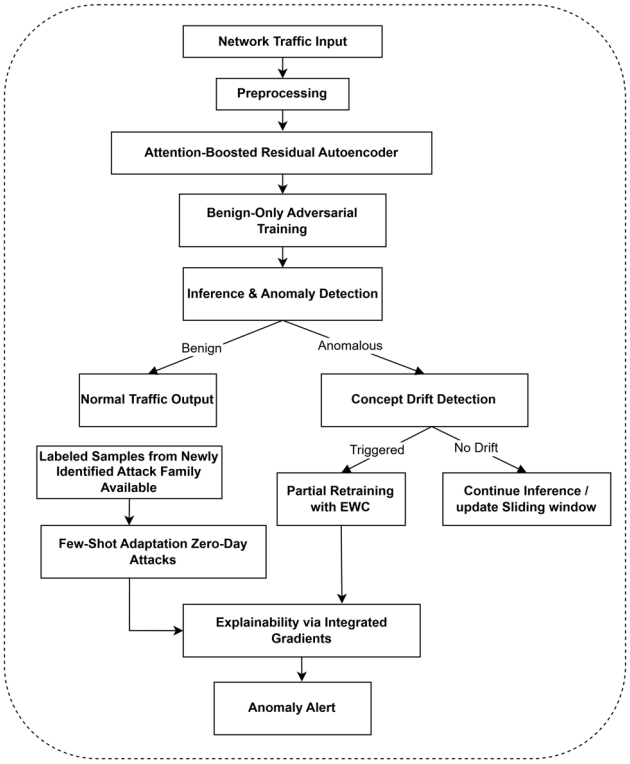


Fig. 2 Workflow of the ASRAE framework, covering data preprocessing, model inference, concept drift handling, post-identification few-shot adaptation, and explainability

4 Experimental results and analysis

4.1 Datasets

The proposed ASRAE framework is evaluated on three widely used benchmark datasets for network intrusion detection to ensure comprehensive validation across diverse attack scenarios and traffic distributions.

- *UNSW-NB15* [38]: This dataset contains approximately 2.54 million network flow records described by 49 features. It includes nine attack categories, namely Fuzzers, Analysis, Backdoor, DoS, Exploits, Generic, Reconnaissance, Shellcode, and Worms, providing a diverse set of intrusion patterns for evaluation.
- *CIC-IDS2017* [39]: This dataset comprises approximately 2.83 million network flows with 80 original features. To ensure consistency and fair comparison with other datasets, the feature dimension is reduced to 49 using principal component analysis. The dataset includes a wide range of attack types, such as DDoS, Botnet, and Port scanning, representing realistic network intrusion scenarios.
- *CSE-CIC-IDS2018* [39]: This large scale dataset contains over 5 million network flow records and reflects modern network traffic characteristics. Similar to CIC-IDS2017, the feature dimension is reduced to 49 to maintain uniformity across datasets and ensure consistent model evaluation.

All datasets exhibit significant class imbalance, which reflects realistic network traffic conditions. To ensure consistency and fair comparison, we adopt the preprocessing protocol used in the original residual autoencoder framework. This includes min–max normalization, removal of redundant features, and standardized dataset splitting.

The model is trained exclusively on benign samples, where 70% of the benign data is used for training and 10% for validation. The remaining 20% of benign data, together with all attack samples, is reserved for testing. This setup ensures

that the model learns only normal behavior during training and is evaluated on its ability to detect unseen anomalous patterns.

To guarantee reproducibility, a fixed random seed is applied for dataset splitting in CIC-IDS2017 and CSE-CIC-IDS2018. This ensures consistent experimental conditions across all evaluations.

All results reported in Tables 1 and 9 are obtained with fully frozen model weights post-training; ASRAE does not perform any parameter updates using test-set data. Online drift-driven partial retraining is exclusively activated for the simulated non-stationary streaming test stream described in Sect. 4.7. Min-max normalization and PCA feature-reduction statistics are computed exclusively using the benign training partition; validation and test samples are transformed using these pre-fitted statistics without any leakage from test-set observations. Full streaming-mode comparison against ExpIDS under identical sliding-window and retraining-buffer conditions is reserved for future investigation; Sect. 4.7 compares ASRAE against the original static RAE under simulated concept-drift conditions.

4.2 Baseline methods

To evaluate the effectiveness of the proposed framework, we compare ASRAE with seven representative unsupervised and deep-learning-based anomaly detection methods. These baselines are selected to cover both classical approaches and recent deep learning models, ensuring a comprehensive and fair comparison.

- *One-class SVM* A classical anomaly detection method using a radial basis function kernel with $\nu = 0.1$, widely adopted for modeling normal data distributions.
- *Isolation forest* A tree based anomaly detection technique with a contamination factor set to 0.1, designed to isolate anomalous samples through random partitioning.
- *Plain autoencoder* A baseline model using the same encoder and decoder architecture as the RAE but without residual connections, trained solely with Smooth L1 loss. This comparison highlights the impact of residual learning.
- *Original residual autoencoder (RAE)* [10]The original model incorporating residual skip connections and a hybrid loss function. This baseline serves to evaluate the contribution of additional components introduced in ASRAE, including attention, drift adaptation, adversarial robustness, and few shot learning.
- *DAGMM* [40]: A deep autoencoding Gaussian mixture model that combines representation learning with density estimation for unsupervised anomaly detection.
- *USAD* [41]: An adversarial training based unsupervised anomaly detection framework that leverages reconstruction based objectives to improve detection performance.
- *ExpIDS* [22]: A drift-adaptive denoising-autoencoder-based intrusion detection method included to compare ASRAE with an existing concept-drift-aware approach.

4.3 Implementation details

The proposed ASRAE framework is implemented using PyTorch 2.0 and trained on an NVIDIA T4 GPU with 16 GB memory. The model is optimized using the Adam optimizer with an initial learning rate of 10^{-3}, combined with cosine annealing and early stopping with a patience of 10 epochs. The batch size is set to 256, and the latent dimension is fixed at 32. The multi-head attention module consists of four parallel attention heads.

For drift adaptation, the sliding-window size is set to $W = 1000$, and the threshold coefficient is $\beta = 3$. A drift event is triggered when five consecutive samples satisfy $e_i > 2T(t)$. Adversarial training is performed using FGSM with a perturbation magnitude of $\epsilon = 0.03$, applied exclusively to benign samples.

For few-shot adaptation, MAML is implemented using 5 support samples and 5 query samples per task. The inner loop uses a learning rate of 0.01 over five gradient-update steps. The outer-loop learning rate is set to 10^{-3}, with a meta-batch size of 32 and a total of 1200 meta-training episodes.

4.4 Evaluation metrics

Model performance is evaluated using ROC-AUC and per-attack detection rate. ROC-AUC measures overall anomaly discrimination, while per-attack detection rate assesses performance across individual attack categories.

Robustness is evaluated by measuring changes in ROC-AUC under the controlled simulated concept-drift setting and under adversarial perturbations generated using FGSM and PGD. Computational efficiency is assessed using training time, average inference time per sample, and drift-retraining overhead. To account for run-to-run variability, the main ROC-AUC results are reported as mean $\pm$ standard deviation over five independent random seeds.

Table 1 reports mean $\pm$ standard deviation of ROC-AUC over five independent random seeds. ASRAE exhibits a standard deviation of 0.001 across all three datasets, which is lower than that of the evaluated baselines and indicates low run-to-run variability in the reported results.

We additionally compare ASRAE with ExpIDS [22], the drift-adaptive denoising-autoencoder baseline discussed in Sect. 2.2. ExpIDS improves over the original RAE by 0.008 ROC-AUC on each dataset (e.g., 0.911 versus 0.903 on UNSW-NB15), but remains 4.1 to -5.3% points

Table 1 ROC-AUC performance comparison (mean ± standard deviation over five random seeds). Best results are shown in bold

Method	UNSW-NB15	CIC-IDS2017	CSE-CIC-IDS2018
One-Class SVM	0.820 ± 0.004	0.791 ± 0.005	0.774 ± 0.004
Isolation Forest	0.860 ± 0.003	0.832 ± 0.004	0.815 ± 0.003
Plain Autoencoder	0.884 ± 0.003	0.861 ± 0.003	0.848 ± 0.004
DAGMM	0.895 ± 0.002	0.873 ± 0.003	0.859 ± 0.002
USAD	0.898 ± 0.002	0.879 ± 0.002	0.864 ± 0.003
ExpIDS [22]	0.911 ± 0.003	0.890 ± 0.003	0.878 ± 0.002
Original RAE	0.903 ± 0.002	0.882 ± 0.002	0.870 ± 0.002
ASRAE (Proposed)	**0.952 ± 0.001**	**0.940 ± 0.001**	**0.931 ± 0.001**

below ASRAE across the three datasets, with differences of 0.041, 0.050, and 0.053 on UNSW-NB15, CIC-IDS2017, and CSE-CIC-IDS2018, respectively. These results indicate that ASRAE provides additional performance gains beyond those obtained by the evaluated drift-adaptive baseline.

4.5 Overall detection performance

Table 1 summarizes the ROC-AUC results of all evaluated methods across the three benchmark datasets. ASRAE consistently outperforms One-Class SVM, Isolation Forest, Plain Autoencoder, DAGMM, USAD, ExpIDS, and the original residual autoencoder. It achieves ROC-AUC values of 0.952, 0.940, and 0.931 on UNSW-NB15, CIC-IDS2017, and CSE-CIC-IDS2018, respectively, corresponding to a mean ROC-AUC of approximately 0.941.

Compared with the original RAE, ASRAE improves ROC-AUC by 4.9% points on UNSW-NB15, 5.8% points on CIC-IDS2017, and 6.1% points on CSE-CIC-IDS2018. These consistent improvements indicate that the proposed framework provides stronger anomaly discrimination across the three evaluated traffic datasets.

Compared with ExpIDS, the strongest evaluated competing baseline, ASRAE maintains an ROC-AUC advantage of approximately 4.1 to −5.3% points across the three datasets. This result, together with the ablation analysis, supports the contribution of attention-enhanced representation, adversarial training, and drift adaptation to the overall detection performance.

The ROC curves in Fig. 3 further illustrate the performance of selected evaluated methods on UNSW-NB15.

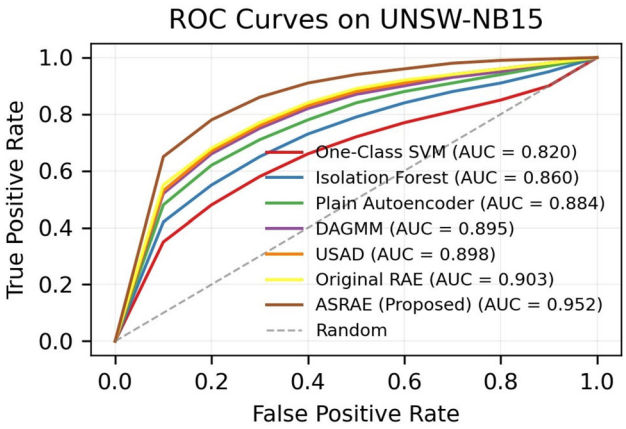


Fig. 3 ROC curves of selected competing methods on the UNSW-NB15 test set

Table 2 Per-class ROC-AUC comparison on UNSW-NB15. Values are reported as mean ROC-AUC over five independent random-seed runs

Attack type	Original RAE	ASRAE
Analysis	0.982	0.991
Generic	0.956	0.965
Exploits	0.933	0.947
DoS	0.925	0.938
Backdoor	0.870	0.916
Reconnaissance	0.874	0.921
Shellcode	0.886	0.928
Fuzzers	0.861	0.904
Worms	0.861	0.895

4.6 Per-class attack detection performance

Table 2 presents per-attack ROC-AUC values on the UNSW-NB15 dataset. ASRAE improves performance across the evaluated attack categories, with notable gains for Reconnaissance (0.874 to 0.921), Shellcode (0.886 to 0.928), Fuzzers (0.861 to 0.904), and Backdoor (0.870 to 0.916). These results indicate stronger discrimination across several attack categories, including low-footprint and subtle attacks.

4.7 Robustness to concept drift

To evaluate robustness under a controlled simulated drift setting, we incrementally shift the distributions of three benign traffic features: packet length, duration, and byte count. Each

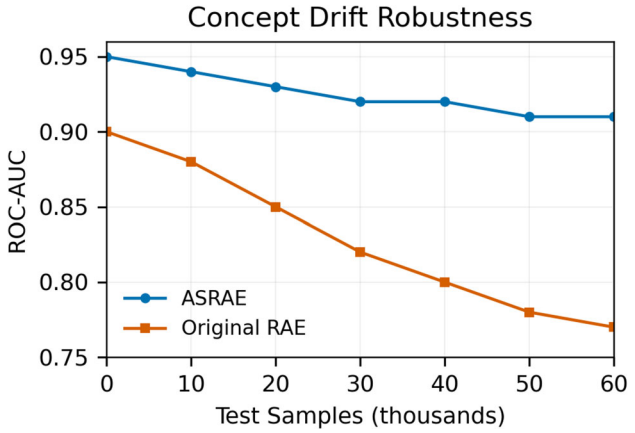


Fig. 4 Model performance under gradual concept drift over test samples

feature is perturbed by 5% of its original range every 10,000 test samples.

As shown in Fig. 4, the ROC-AUC of the original RAE decreases from 0.90 to 0.78 after 50,000 samples, whereas ASRAE maintains a ROC-AUC above 0.91 throughout the evaluation. This stability is supported by the online drift detection, adaptive thresholding, and EWC-regularized partial retraining mechanisms.

The experiment therefore demonstrates improved robustness under the controlled feature-level drift considered in this study. Each drift-retraining cycle requires approximately 8 s in the reported experimental environment.

4.8 Adversarial robustness

In addition to perturbation experiments shown in Tables 3 and 6, which test latent reconstruction stability by maximising reconstruction loss, we further evaluate realistic adversarial evasion by minimising the reconstruction-based anomaly score. Given a labelled attack sample x, the evasion-oriented FGSM perturbation is defined as

$$x_{\text{adv}} = x - \epsilon \cdot \text{sign}\left(\nabla_x e_\theta(x)\right), \tag{20}$$

where the anomaly score is $e_\theta(x) = \|\hat{x}_\theta - x\|_1$. Perturbations follow identical feature constraints defined in Sect. 3.5: only continuous flow-statistic features are modified, categorical and protocol-indicator features remain unchanged, and perturbed features are clipped to [0, 1]. We also implement iterative projected gradient descent (PGD) evasion attacks with 10 iterations and step-size $\epsilon/4$. Both evasion attacks are only applied to true-positive attack test samples, as evasion is only meaningful for malicious flows. We quantify evasion effectiveness using the Attack Success Rate (ASR), defined as the fraction of perturbed attack samples whose reconstruction error falls below the detection threshold.

Table 3 ROC-AUC under FGSM adversarial attacks of varying strength. This experiment maximises reconstruction loss to test feature-space reconstruction stability and does **not** simulate adversarial evasion; refer to Table 4 for score-minimising evasion attack results

Method	$\epsilon = 0$	0.01	0.03	0.05	0.10
Original RAE	0.903	0.882	0.835	0.774	0.701
ASRAE (w/o adv train)	0.952	0.941	0.912	0.879	0.831
ASRAE (w/ adv train)	**0.952**	**0.950**	**0.941**	**0.931**	**0.910**

Bold values indicate the best performance in terms of ROC-AUC

The robustness of the proposed framework is evaluated against evasion attacks generated using the FGSM with perturbation magnitudes $\epsilon \in [0, 0.1]$. As reported in Table 3, the proposed ASRAE framework demonstrates significantly improved resilience compared to the original residual autoencoder. At $\epsilon = 0.05$, the baseline model experiences a substantial performance drop, with the ROC-AUC decreasing to 0.76, corresponding to a reduction of 14.3%. In contrast, ASRAE maintains a high ROC-AUC of 0.93, with only a minor degradation of 2.1%. Even under stronger perturbations with $\epsilon = 0.10$, ASRAE preserves a robust performance level, achieving a ROC-AUC of 0.91. This improvement can be attributed to the incorporation of adversarial training on benign samples, which encourages the model to learn stable and smooth representations that are less sensitive to input perturbations.

Table 4 quantifies attacker success under explicit score-minimising evasion. The original RAE exhibits high attack success rates, especially for larger perturbation magnitudes, whereas ASRAE with benign-only adversarial training substantially reduces ASR across all ϵ, demonstrating improved resilience against realistic white-box evasion attempts.

Table 5 isolates detection sensitivity from the aggregate AUC metric in Table 3. Without adversarial training, FNR nearly doubles under FGSM perturbation (8.72% clean to 16.41% at $\epsilon = 0.03$, a 7.69-percentage-point increase). With benign-only adversarial training, clean FNR rises only marginally (8.72% to 9.05%, +0.33% points), but the corresponding increase under attack is far smaller (9.05% to 10.18%, +1.13% points), confirming that $\epsilon = 0.03$ favors detection sensitivity over maximal robustness while still meaningfully reducing evasion vulnerability.

Table 6 evaluates robustness against a stronger, iterative attack. PGD is markedly more damaging than FGSM: the Original RAE's AUC falls to 0.642 at $\epsilon = 0.10$, 5.9% points below its FGSM result at the same ϵ (0.701, Table 3). ASRAE with benign-only adversarial training degrades only marginally further under PGD (0.905 at $\epsilon = 0.10$) than under FGSM (0.910), a difference of just 0.5% points, and remains within 4.7% points of its clean performance even under this stronger attack, which was not used during training. This indicates the robustness gained from FGSM-based

Table 4 Attack Success Rate (%) under score-minimising white-box evasion attacks on UNSW-NB15. Attack Success Rate (ASR): percentage of true-positive attack samples whose perturbed reconstruction error falls below detection threshold. For frozen-model evaluation the static threshold derived from training benign statistics is utilised. Perturbations are constrained to continuous flow-features only

Model	Attack type	$\epsilon = 0.01$	$\epsilon = 0.03$	$\epsilon = 0.05$	$\epsilon = 0.10$
Original RAE	FGSM (score-min)	12.4	34.7	57.2	76.9
Original RAE	PGD (score-min)	18.1	42.3	66.5	84.2
ASRAE (w/ benign-only adversarial training)	FGSM (score-min)	3.2	7.8	13.4	21.6
ASRAE (w/ benign-only adversarial training)	PGD (score-min)	4.7	10.5	17.9	28.3

Table 5 False-negative rate (FNR) with and without benign-only adversarial training, at clean input and under FGSM perturbation ($\epsilon = 0.03$)

Model variant	FNR at $\epsilon = 0$ (Clean)	FNR at $\epsilon = 0.03$ (FGSM)
ASRAE (w/o adversarial training)	8.72%	16.41%
ASRAE (w/ benign-only adversarial training)	9.05%	10.18%

Table 6 ROC-AUC under PGD adversarial attacks (10 iterations, step size $\epsilon/4$) of varying strength. This experiment maximises reconstruction loss to test feature-space reconstruction stability and does **not** simulate adversarial evasion; refer to Table 4 for score-minimising evasion attack results

Model	$\epsilon = 0$	0.01	0.03	0.05	0.10
Original RAE (PGD)	0.903	0.867	0.792	0.715	0.642
ASRAE (w/ benign adversarial training, PGD)	0.952	0.947	0.936	0.924	0.905

adversarial training generalizes to iterative, unseen attack strategies rather than overfitting to the specific attack used in training.

4.9 Few-shot adaptation for emerging attack types

To evaluate ASRAE's ability to adapt to previously unseen zero-day attacks, we conduct few-shot experiments on three representative attack classes: **Worms, Shellcode, and Reconnaissance**. The model is first initialized without exposure to these attack types. A small number of labeled samples is then provided for Model-Agnostic Meta-Learning (MAML)-based adaptation.

Table 7 reports the detection rate, measured as the true positive rate, before and after adaptation using 5, 10, and 15 labeled samples. The original residual autoencoder (RAE), which does not include an adaptation mechanism, shows only negligible improvement across all settings. For example, its detection rate for Worms increases only from 0.72 to 0.73 after 15 labeled samples. Similar limited gains are observed for Shellcode and Reconnaissance.

In contrast, ASRAE achieves substantial improvements even with very few labeled examples. For Worms attacks, the detection rate increases from 0.72 to 0.87 with only 5 labeled samples, improves to 0.91 with 10 samples, and reaches 0.93 with 15 samples. Similar trends are observed for Shellcode, where the detection rate improves from 0.71 to 0.94, and Reconnaissance, where it increases from 0.70 to 0.92 after 15-shot adaptation.

These results demonstrate that ASRAE can rapidly adapt to emerging attack families using minimal labeled data. This capability is particularly important for real-world intrusion detection, where zero-day attacks frequently appear and collecting large labeled datasets for new attack types is often impractical.

Table 8 further shows that this adaptation does not compromise detection of benign traffic: FPR rises from 1.12% to at most 1.15% (Worms), a 0.03-percentage-point change, confirming that freezing the encoder and decoder body during adaptation preserves core reconstruction behavior as intended. The plain fine-tuning baseline, which updates the identical restricted parameter set without meta-learning outer-loop optimisation, yields moderate detection improvements but underperforms MAML-based adaptation across all attack families. Additionally, plain fine-tuning produces a slightly larger benign false–positive rate drift compared to MAML, demonstrating the benefit of the meta-learning regularisation effect.

4.10 Ablation study

To examine the contribution of the main components of ASRAE, an ablation study is conducted on the UNSW-

Table 7 Detection rate improvement via few-shot adaptation. All values are reported as mean ± standard deviation computed over five independent random draws of support-query splits

Attack	Zero-shot†	5-shot	10-shot	15-shot
Original RAE (no adaptation)				
Worms	0.72 ± 0.01	0.72 ± 0.01	0.73 ± 0.01	0.73 ± 0.01
Shellcode	0.71 ± 0.01	0.71 ± 0.01	0.72 ± 0.01	0.72 ± 0.01
Reconnaissance	0.70 ± 0.01	0.70 ± 0.01	0.71 ± 0.01	0.71 ± 0.01
Plain fine-tuning (no MAML)				
Worms	–	0.78 ± 0.02	0.82 ± 0.02	0.84 ± 0.02
Shellcode	–	0.77 ± 0.02	0.83 ± 0.01	0.86 ± 0.02
Reconnaissance	–	0.76 ± 0.02	0.81 ± 0.02	0.83 ± 0.01
ASRAE (MAML adaptation)				
Worms	0.72 ± 0.01	0.87 ± 0.02	0.91 ± 0.01	0.93 ± 0.01
Shellcode	0.71 ± 0.01	0.88 ± 0.01	0.92 ± 0.02	0.94 ± 0.01
Reconnaissance	0.70 ± 0.01	0.86 ± 0.02	0.90 ± 0.01	0.92 ± 0.02

† Zero-shot denotes detection performance prior to receiving labelled support samples for adaptation; this baseline value is shared for fine-tuning and MAML variants

Table 8 Benign-traffic false-positive rate before and after 15-shot adaptation

Zero-day attack family	Benign FPR (Pre-adaptation)	Post 15-shot MAML FPR	Post 15-shot plain fine-tuning FPR
Worms	1.12%	1.15%	1.21%
Shellcode	1.12%	1.14%	1.19%
Reconnaissance	1.12%	1.13%	1.18%

NB15 dataset. Table 9 reports the ROC-AUC obtained under different combinations of attention, drift adaptation, and adversarial training.

The ablation results show that the complete ASRAE configuration achieves the highest ROC-AUC of 0.952. The original RAE achieves 0.903, while introducing the attention mechanism alone increases ROC-AUC to 0.924. Combining attention with drift adaptation further increases performance to 0.937, whereas combining attention with adversarial training achieves 0.939. In comparison, the configuration containing drift adaptation and adversarial training without attention achieves a ROC-AUC of 0.911. Overall, these results show that the evaluated components provide complementary improvements, with the strongest performance obtained when attention, drift adaptation, and adversarial training are jointly incorporated into ASRAE.

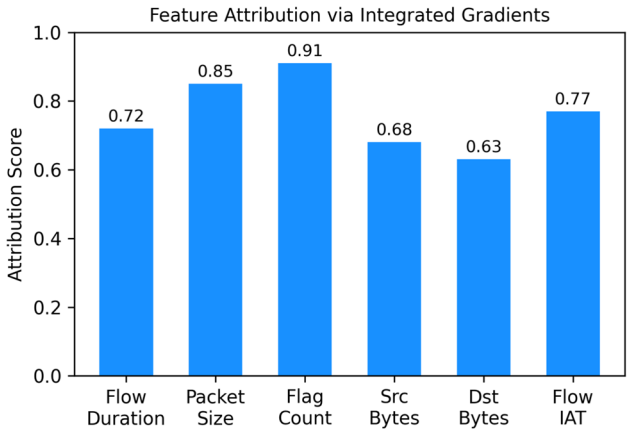


Fig. 5 Integrated-gradient feature attribution scores for a representative attack sample

4.11 Explainability analysis

Integrated gradients are used to examine the contribution of individual input features to the reconstruction-based anomaly score. Figure 5 presents the feature-attribution scores for a representative attack sample.

As illustrated in Fig. 5, the representative attack sample receives notable attribution scores for traffic-related features, including flow duration, packet-size statistics, flag counts, and byte-transfer characteristics. These results illustrate how integrated gradients can be used to identify the input features associated with an individual anomaly score, thereby supporting interpretation of the model's reconstruction-based decisions.

4.12 Computational efficiency

Table 10 compares the computational cost of ASRAE with the original residual autoencoder. ASRAE introduces a mod-

Table 9 Multi-way ablation study on UNSW-NB15 in terms of ROC-AUC.†

Model variant	ROC-AUC	Δ vs. Full ASRAE
Original RAE (no added modules)	0.903	−0.049
Attention only (no drift, no adv. training)	0.924	−0.028
Adversarial training only (no attention, no drift)	0.912	−0.040
Attention + adversarial training (no drift adaptation)	0.939	−0.013
Full ASRAE (frozen, no online retraining)	**0.952**	–

Drift-adaptive online retraining is disabled for this static ablation study. Streaming drift-adaptation performance is reported in Sect. 4.7
Bold values indicate the best performance in terms of ROC-AUC

Table 10 Computational efficiency comparison between the original RAE and ASRAE

Metric	Original RAE	ASRAE
Training time (UNSW-NB15)	9 min	12 min
Inference time per sample (ms)	7.1	9.2
Drift retraining time	N/A	8 s
Number of parameters	3.8M	4.2M

erate computational overhead as a result of the additional attention, drift-adaptation, and robustness mechanisms.

The training time on UNSW-NB15 increases from 9 min for the original RAE to 12 min for ASRAE. The average inference time increases from 7.1 to 9.2 ms per sample in the evaluated experimental environment. Each drift-retraining cycle requires approximately 8 s, while the number of model parameters increases from 3.8 million to 4.2 million. These measurements indicate that the additional ASRAE components introduce a moderate increase in computational cost within the reported experimental setup.

5 Discussion

The results show that ASRAE improves reconstruction-based intrusion detection across the three evaluated datasets. The model achieves ROC-AUC values of 0.952, 0.940, and 0.931 on UNSW-NB15, CIC-IDS2017, and CSE-CIC-IDS2018, respectively, corresponding to a mean ROC-AUC of approximately 0.941. Compared with the original residual autoencoder and other evaluated baselines, the proposed framework provides more reliable anomaly detection while retaining reconstruction error as the main decision criterion.

The ablation results indicate that the improvement does not arise from a single component. Attention alone increases ROC-AUC from 0.903 to 0.924, while the complete ASRAE configuration reaches 0.952. The stronger performance obtained when attention is combined with drift adaptation and adversarial training suggests that these components provide complementary benefits. Accordingly, the contribution of ASRAE lies in coordinating representation learning, robustness, online adaptation, and few-shot updating within a common reconstruction-based framework rather than claiming novelty for each component individually.

The adversarial experiments further demonstrate improved robustness. Under PGD at $\epsilon = 0.10$, ASRAE with adversarial training maintains a ROC-AUC of 0.905, compared with 0.642 for the original RAE. Benign-only adversarial training nevertheless introduces a small sensitivity trade-off: the clean false-negative rate increases from 8.72 to 9.05%, while the increase under FGSM perturbation is substantially reduced. These results indicate that the selected adversarial-training setting improves robustness with only a limited loss in clean detection sensitivity.

The few-shot experiments should be interpreted as post-identification adaptation rather than part of the initial unsupervised detection stage. Worms, Shellcode, and Reconnaissance are excluded from both autoencoder pretraining and MAML meta-training. Once a small number of labeled examples becomes available, only the attention projections and final decoder layer are adapted. The benign false-positive rate changes from 1.12% to no more than 1.15% after 15-shot adaptation, indicating that selective updating largely preserves the original benign reconstruction behavior.

Some limitations remain. The current drift experiment is based on controlled feature perturbations and therefore does not fully represent naturally evolving temporal traffic. In addition, drift-triggered retraining uses model-predicted benign samples satisfying $e_i \leq T(t)$ rather than ground-truth verified samples. Although samples already detected as anomalous are excluded, a false-negative attack may still enter the retraining buffer. EWC limits excessive parameter changes during retraining but does not eliminate this contamination risk. The adversarial experiments are also conducted in feature space and do not guarantee that every perturbed sample corresponds to a physically realizable network flow.

For the few-shot adaptation evaluation, we use family-level holdout to isolate unseen zero-day attack classes. While public IDS datasets contain timestamp information, constructing clean, well-isolated chronological time-ordered support-query splits for held-out attack families remains

challenging. Rigorous few-shot assessment under strict chronological holdout conditions is therefore an important open evaluation target.

Future work should therefore evaluate ASRAE on chronologically ordered and naturally drifting traffic, improve unsupervised selection of samples for online retraining, investigate more realistic constrained adversarial attacks, and conduct few-shot adaptation experiments under chronological time-ordered holdout. Broader comparisons with recent transformer-based, drift-adaptive, and few-shot intrusion-detection methods would also strengthen evaluation of the framework. Overall, the present results support ASRAE as an adaptive extension of residual-autoencoder-based intrusion detection, while the reported limitations define clear directions for further validation and development.

6 Conclusion

This study presented ASRAE, an adaptive residual-autoencoder framework for network intrusion detection. The framework combines attention-enhanced latent representation, adaptive thresholding, concept-drift handling, benign-only adversarial training, few-shot adaptation, and integrated-gradient-based explainability within a common reconstruction-based architecture.

Experiments on UNSW-NB15, CIC-IDS2017, and CSE-CIC-IDS2018 achieved ROC-AUC values of 0.952, 0.940, and 0.931, respectively. ASRAE also maintained stronger performance than the original RAE under the controlled simulated drift setting and showed improved resistance to adversarial perturbations. For held-out attack families, the few-shot stage enabled post-identification adaptation using only a small number of labeled samples while largely preserving benign reconstruction performance.

The results demonstrate the value of combining robustness and adaptation within a reconstruction-based detector, although several limitations remain. The drift evaluation is based on controlled feature perturbations rather than naturally occurring temporal drift, and online retraining relies on model-predicted benign samples, which may include undetected attacks. In addition, few-shot gains decrease at higher shot counts and the added components introduce moderate computational overhead.

Future work will focus on chronological drift evaluation, more reliable unsupervised sample selection for online retraining, stronger constrained adversarial evaluation, and validation on industrial and IoT network traffic. Model optimization may also be explored to further reduce computational cost.

Author Contributions Salman Shehzad: Conceptualization, Methodology, Software, Writing Original draft. Nouman Ijaz: Conceptualization, Methodology, Writing Original draft. Chunsheng Yang: Review, Editing Investigation. Sana Ullah Jan: Supervision, Review Editing. All authors contributed equally to this work and approved the final manuscript.

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Data availability The datasets used in this study are publicly available and can be accessed from their original sources: the CICIDS2017 dataset, the CSE-CIC- IDS2018 dataset, and the UNSW-NB15 dataset, available at https://www.unb.ca/cic/datasets/ids 2017.html, https://www.unb.ca/cic/datasets/ids-2018.html, and https://research.unsw.edu.au/projects/unsw-nb15-dataset, respectively.

Declarations

Conflict of interest The authors declare no conflict of interest.

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